Threebit Hacker
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In order to join or initiate a conversation, participants need to exchange cryptographic key material. To address this problem, we created a slightly unusual design: *contact vouchers*.

In many systems, invites to conversations flow from an existing member of the conversation to the user being invited. In our contact voucher protocol, this flow is reversed: Bob, who wishes to join a conversation, hands a contact voucher (out-of-band) to existing member Alice, who then inducts Bob into the group.

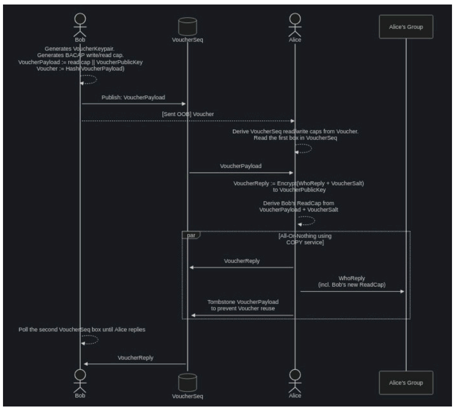
This design mitigates two potential problems with the conventional way of doing things:

1. A third party who observes the contact voucher does not get to read either participant's actual messages. However,
   - A **passive** adversary learns that the voucher was spent, but does not get to observe further interactions.
   - An **active** adversary can create a new fake group to induct the member into, but does not learn anything about the existing group.

   In the future, to prevent this one-way impersonation, we could require that Alice and Bob both bring something on paper to their meeting.
   - Bob brings the contact voucher.
   - Alice brings a fingerprint for the `VoucherReplyPublicKey` (thwarting the active attacker).
2. Only one thing that needs to delivered out-of-band to achieve a two-pass protocol (instead of a three-pass protocol):

   - One of the parties needs to bring key material to a meeting in order to establish contact.

The following diagram illustrates the contact voucher authentication handshake.



Self-authenticating BACAP payload

The first message sent, the `VoucherPayload`, is authenticated in the following manner:

- The `VoucherPayload` is computed (first).
- A cryptographic hash of the `VoucherPayload` is computed. This hash **is** the `Voucher`.
- The `Voucher` is then used to derive a BACAP read/write capability set.
- The `VoucherPayload` is uploaded to the sequence described by the capability (at index 0).
- Anyone who intercepts the `Voucher` can read from **and** write to the message sequence.
- But: Since the `Voucher` is a hash over the `VoucherPayload`, writing the sequence with anything but the `VoucherPayload` will be detectable by the recipient.
- This means that the contents *cannot* be undetectably modified by the interceptor.